

The sign multiplier destroys the adapted one-molecule condition

A counterexample to Remark 7.3 of arXiv:1403.5367

Candidate solution packet

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Status. Candidate full negative answer to the concrete uncertainty in Remark 7.3, likely valid pending expert review.

1 Source question

Auscher and Stahlhut work with a constant-coefficient first-order operator D , an accretive multiplication operator B , and Hardy spaces associated with the bisectorial operators DB and BD . In their perturbation argument, an adapted atom $a = Du$ is sent by a bounded functional-calculus multiplier to

$$m = b(DB)a = Dv.$$

Their estimates show that m is a classical Hardy-space molecule, but the available bound for its potential v loses an extra annular factor 2^j . Remark 7.3 asks whether m is nevertheless a single $(\mathbb{H}_D^p, \varepsilon, 1)$ -molecule in their adapted sense [1].

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Remark 7.3. It is unclear to us whether m is itself an $(\mathbb{H}_D^p, \varepsilon, 1)$ -molecule in the sense of our definition. One can indeed write $m = Dv$ with $v = \int_0^\infty (b\psi_t)(BD)u \frac{dt}{t}$ and obtain by the same method

$$\|v\|_{L^2(S_j(Q))} \lesssim \|b\|_\infty (2^j \ell(Q))^{\frac{n}{2} - \frac{n}{p}} 2^{-j\varepsilon} 2^j.$$

There is an extra factor 2^j . However, this is sufficient to prove a uniform L^{p^*} bound on v if one needs it.

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This section will be used in Section 14 below

Figure 1: Remark 7.3 on source PDF page 45.

For the case used below, $n = p = 1$, the relevant part of the definition says that a normalized $(\mathbb{H}_D^1, \varepsilon, 1)$ -molecule m associated with an interval I of length $\ell(I)$ has a potential $v \in \text{Dom } D$ such that $Dv = m$ and

$$\|(\ell(I)D)^{-1}m\|_{L^2(S_j(I))} \leq (2^j \ell(I))^{-1/2} 2^{-j\varepsilon} \quad (j \geq 0), \quad (1)$$

where $S_j(I) = 2^j I \setminus 2^{j-1} I$ for $j \geq 1$. Allowing a fixed multiplicative normalization constant would make no difference below.

2 Proof intuition

The multiplier that most directly exposes the obstruction is the sign of the derivative. On the Fourier side, $D = -i d/dx$ has symbol ξ , while $\operatorname{sgn}(D)$ is, up to a constant, the Hilbert transform. Applying it to $a = Du$ gives a potential

$$v = \operatorname{sgn}(D)u.$$

If the compactly supported u has nonzero integral, the Hilbert transform has the unavoidable far-field term c/x . Its L^2 mass on an annulus of radius R is therefore comparable from below to $R^{-1/2}$. The positive ε in the adapted molecule definition demands the strictly faster rate $R^{-1/2-\varepsilon}$. Since D has no nonzero L^2 kernel, one cannot choose a different potential to remove this tail.

3 Counterexample

Theorem 1. *In the admissible model*

$$n = N = 1, \quad D = -i \frac{d}{dx} \text{ on } L^2(\mathbb{R}), \quad B = I,$$

there are an $(\mathbb{H}_D^1, 1)$ -atom a and a function $\beta \in H^\infty(S_\mu)$ such that $m = \beta(DB)a$ is not an $(\mathbb{H}_D^1, \varepsilon, 1)$ -molecule for any $\varepsilon > 0$. Indeed, m is not such a molecule relative to any interval. Consequently, the question in Remark 7.3 has a negative answer.

Proof. Let $Q = (-1, 1)$. Choose $\varphi \in C_c^\infty((-1/2, 1/2))$ with

$$M_0 := \int_{\mathbb{R}} \varphi(y) dy \neq 0.$$

For a sufficiently small nonzero scalar c , put $u = c\varphi$ and $a = Du$. Then a and u are supported in Q , and scaling c makes

$$\|a\|_2 \leq |Q|^{-1/2}, \quad \|u\|_2 \leq \ell(Q)|Q|^{-1/2}.$$

These are precisely the two normalization estimates for an $(\mathbb{H}_D^1, 1)$ -atom in the source's definition. Notice that $\int u = cM_0 \neq 0$.

The bisector S_μ has two connected components. Define

$$\beta(z) = \begin{cases} 1, & z \in S_{\mu,+}, \\ -1, & z \in S_{\mu,-}. \end{cases}$$

This is a bounded holomorphic function on S_μ , with $\|\beta\|_\infty = 1$. The Fourier symbol of D is ξ , so $\beta(D)$ is the multiplier $\operatorname{sgn} \xi$. Set

$$m = \beta(D)a.$$

Functional calculus commutes with D on its domain; therefore

$$m = \beta(D)Du = D\beta(D)u. \tag{2}$$

Put $v = \beta(D)u$. Since the L^2 kernel of D is zero, equation (2) also shows that v is the *unique* L^2 potential of m .

With the convention

$$Hf(x) = \frac{1}{\pi} \text{p. v.} \int_{\mathbb{R}} \frac{f(y)}{x-y} dy,$$

the Hilbert transform has Fourier multiplier $-i \operatorname{sgn} \xi$. Thus $v = iHu$. For $|x|$ larger than twice the support radius of u , no principal value is needed and

$$v(x) = \frac{i}{\pi} \int_{\mathbb{R}} \frac{u(y)}{x-y} dy \quad (3)$$

$$= \frac{i}{\pi x} \int_{\mathbb{R}} u(y) dy + O(|x|^{-2}). \quad (4)$$

The coefficient of x^{-1} is nonzero. Hence there are $R_0, c_0 > 0$ such that

$$|v(x)| \geq \frac{c_0}{|x|} \quad (x \geq R_0). \quad (5)$$

Suppose, toward a contradiction, that m is an $(\mathbb{H}_D^1, \varepsilon, 1)$ -molecule for some $\varepsilon > 0$, associated with some fixed interval I . By uniqueness, the potential occurring in its definition must be the above v . For all sufficiently large j , the right component of $S_j(I)$ contains an interval of length comparable to $R_j = 2^j \ell(I)$ on which x is comparable to R_j . From (5),

$$\|v\|_{L^2(S_j(I))} \geq c_I R_j^{-1/2}. \quad (6)$$

On the other hand, multiplying (1) by $\ell(I)$ gives

$$\|v\|_{L^2(S_j(I))} \leq \ell(I) R_j^{-1/2} 2^{-j\varepsilon}. \quad (7)$$

The ratio of the lower bound in (6) to the upper bound in (7) grows like $2^{j\varepsilon}$. This is impossible as $j \rightarrow \infty$. Since both I and $\varepsilon > 0$ were arbitrary, m is not a first-order adapted molecule relative to any interval. $\square$

4 What the example shows

The failure is specifically in the potential estimate ($k = 1$), not in the ordinary decay of m . Differentiating (3) gives a $|x|^{-2}$ tail for $m = Dv$, consistent with the source's conclusion that m is a classical H^1 molecule. The primitive necessarily loses one power, exactly the low-frequency phenomenon reflected by the source's extra 2^j factor. Thus that loss cannot be removed in the stated generality.

The counterexample does not address the memoir's broader questions about endpoint functional calculus, abstract atomic decompositions, or stability at all coincidence exponents.

5 Verification and novelty status

The proof was checked against the source definitions at four points: the normalization of an adapted atom; admissibility of a function taking separate constants on the two components of a bisector; uniqueness of the L^2 D -potential; and the factor $\ell(I)$ obtained from $(\ell(I)D)^{-1}m$. The tail calculation is an explicit expansion of the Hilbert-transform kernel and uses no external theorem.

On August 11, 2026, the local run indexes were searched by arXiv id, exact title, Remark 7.3 terminology, and the adapted-molecule phrase. Exact-phrase web searches found copies of the source memoir but no later answer, and sign/Hilbert-transform searches found no matching resolution. This was a bounded search, so novelty confidence is moderate pending a specialist citation review.

References

- [1] P. Auscher and S. Stahlhut, *A priori estimates for boundary value elliptic problems via first order systems*, Part I of *Functional Calculus for First Order Systems of Dirac Type and Boundary Value Problems*, Mém. Soc. Math. Fr. (N.S.) **144** (2016); arXiv:1403.5367.